

EXERCISE 1: HYPOTHESIS TESTING

1. StatCrunch for statistics

For effective learning, an instructor advised his students to practice solving statistical problems on StatCrunch, a web-based software. In an in-class activity, a student surveyed 15 of her 35 classmates to assess the number of hours they spend per week practicing statistics on StatCrunch. The following data was collected:

6.5 4 3.5 0 12 5 12 15 12 0 1 8 0.5 2 7

At the 0.05 level of significance, is there strong evidence that the mean number of hours of StatCrunch usage in the entire class is different from 7? Answer by:

- Identifying the relevant variable and parameter.
- Stating null and alternative hypotheses.
- Finding and interpreting the test statistic value.
- Reporting and interpreting the P-value and stating the conclusion in context.

Source (Data): Agresti, A., Franklin, C. and Klingenberg, B. (2018). Statistics: The Art and Science of Learning from Data, 4th Edition, Global Edition. Pearson.

2. The waiting time

A bank branch located in a commercial district of a city has the business objective of improving the process for serving customers during the noon-to-1:00 P.M. lunch period. The waiting time (defined as the time the customer enters the line until he or she reaches the teller window) of a random sample of 15 customers is collected. These data are:

4.21 5.55 3.02 5.13 4.77 2.34 3.54 3.20 4.50 6.10 0.38 5.12
6.46 6.19 3.79

- At the 0.05 level of significance, is there evidence that the population mean waiting time is less than 5 minutes?
- What assumption about the population distribution is needed in order to conduct the t test in (a)?

- c. Construct a boxplot or normal probability plot to evaluate the assumption made in (b).
 - d. Do you think that the assumption needed in order to conduct the t test in (a) is valid? Explain.
 - e. As a customer walk into the branch office during the lunch hour, she asks the branch manager how long she can expect to wait. The branch manager replies, "Almost certainly less than 5 minutes."
- On the basis of results of (a), evaluate this statement.

Source: Berenson, M., Levine, D.M., and Krehbiel, T.C. (2012). *Basic Business Statistics: Concepts and Applications*, 12th edition, Pearson.

3. The strength of the insulators

A manufacturing company produces electrical insulators. If the insulators break when in use, a short circuit is likely to occur. To test the strength of the insulators, destructive testing is carried out to determine how much force is required to break the insulators. Force is measured by observing the number of pounds of force applied to the insulators before it breaks. The following data are from 30 insulators subjected to this testing:

1,870	1,728	1,656	1,610	1,634	1,784	1,522	1,696	1,592	1,662	1,866
1,764	1,734	1,662	1,734							
1,774	1,550	1,756	1,762	1,866	1,820	1,744	1,788	1,688	1,810	1,752
1,680	1,810	1,652	1,736							

- a. At the 0.05 level of significance, is there evidence that the population mean force required to break the insulator is greater than 1,500 pounds?
- b. What assumption about the population distribution is needed in order to conduct the t test in (a)?
- c. Construct a boxplot or normal probability plot to evaluate the assumption made in (b).
- d. Do you think that the assumption needed in order to conduct the t test in (a) is valid? Explain.

Source: Berenson, M., Levine, D.M., and Krehbiel, T.C. (2012). *Basic Business Statistics: Concepts and Applications*, 12th edition, Pearson.